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EDGE COMPUTING

Multi-Access Edge Computing will enable new business opportunities for Telcos

Abstract:

Telecom Service Providers are slowly transforming into Digital Service Providers in order to stay relevant to the customer and to ensure that they can reap the profits of the increasing digital economy. To stave off competition from hyperscalers such as Google, Amazon, Facebook and Microsoft, Telcos are turning to new age applications and services. Telcos are uniquely positioned to take advantage of their physical proximity to the customer to provide compute and storage and new digital services. This will require that Telcos work with a multitude of partners resulting in engendering a new ecosystem that fosters agile service delivery which is cloud native and pay-as-you-go business model.

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Introduction

The telecom industry is at the cusp of a paradigm shift. While 2G focused on calling and messaging, 3G brought with it the basic ability to access webpages and email. 4G provided the ability to transmit video. The process of commoditization of connectivity began with 4G. With 5G, garnering revenues by just providing connectivity will become a passé. According to latest [market research reports](#), Mobile operator voice revenue is likely to drop to \$208B by 2024 from \$381B in 2019, as users continue to prefer OTT (over-the-top) services.

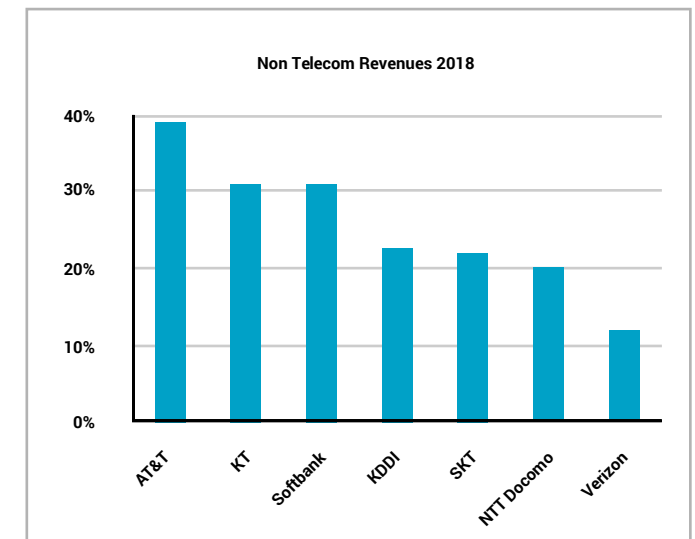
Telecom operators (Telcos) and Communication service providers (CSPs) are transforming from connectivity providers to digital service providers to avoid commoditization and garner more revenue. For example, Jio has digital offerings such as Saavn, JioMart, and JioMeet while Airtel has Wynk, Nxtra, Xstream, and Verizon has Digital Media, Traffic Management, etc. With 5G, this trend will deepen further as the battle shifts

to consumer data, apps, and enterprise services. For example, AT&T garnered nearly 40% of its revenue from non-telecom sources in 2018 (Source: GSMA 2020).

Digital solutions need to be provided as a service and on demand; Cloud/Data Centers fit the bill. As a result, we see a steady 'cloudification' of the Telcos. The IT capex spend is [forecasted to reach nearly \\$40B](#) by 2025 from \$8B in 2019 by virtue of increased virtualization/softwarization of telecom equipment.

The virtualization and disaggregation of the telecom network has brought about a great disruption for the players in the industry. Virtualization decouples the software from hardware, enabling the use of commercial servers in the network, thus reducing capex for telcos. Other applications can run on the same infrastructure to provide value added services, such as video optimization, caching, and localization. Telcos are now offering IT cloud environments within the access networks, enabling

the multiplication of the application ecosystem. This paper talks about how Multi-Access Edge Computing (MEC) a.k.a. Edge Cloud or Edge Computing is spawning new applications and reshaping the ecosystem, what is shaping the Edge Cloud and how it impacts the telcos' own fortunes, having lost the initial battle to OTT apps and cloud service providers.



Source: GSMA 2020



Decentralization of Telco Architecture – Data Center in the RAN

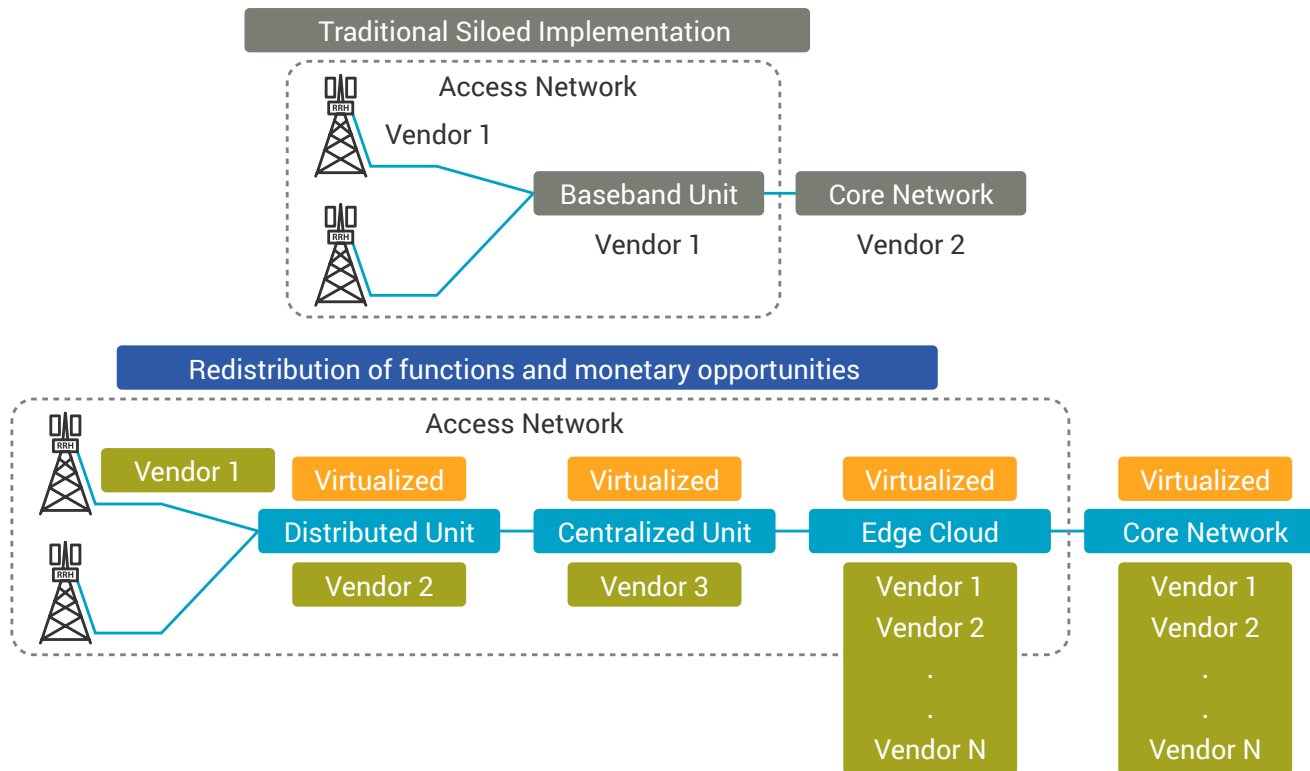


Figure 1: Traditional and Disaggregated Networks

a) Edge Cloud

The digitalization of the telco network is providing a fillip to the applications ecosystem. Newer applications are emerging which are also demanding a lot from the network. For example, as content (video and otherwise) consumption increases, it is laying a lot of stress on the fiber backbone. To meet the bandwidth demand, CSPs must invest in fiber which is very expensive. Traditionally, content is fetched from a content delivery network (CDN) to the centralized telco cloud and then relayed to the user through the access network.

If there is video that is going “viral”, this content must be repeatedly fetched from the CDN to cater to 100s of thousands, if not millions, of users within a short time (ranging from a few hours to a few days), thus clogging up the fiber backhaul. Similarly, popular TV shows from Amazon and Netflix have to be streamed to a very large audience. If the popular content can be stored locally within the access network of the telco, the data need not be repeatedly hauled from CDNs, hence freeing up bandwidth.



Another interesting use case can be that of thousands of users receiving [live video stream from cameras in a stadium in real time](#). The user can choose the camera angle and view replays. Thus, all media is produced and consumed locally without having to change any of the core network elements.

There are [applications such as AR/VR and C-V2X](#) which require extremely low latency which the traditional networks cannot meet. For AR applications the precise location of the user should be known so that the image

is refreshed every time the user moves. Such information processing needs to happen at a fast rate. Performing computations and data transfer from a centralized cloud located several hundreds of kilometers from the user will not work. It must be done very close to the end user instead.

The available channel bandwidth may vary significantly based on the channel conditions as well as the number of devices entering and leaving the network at any given time. The channel bandwidth can vary by an

order of magnitude within a matter of seconds. [TCP may not be able to adapt fast enough to rapidly varying conditions in the radio access network \(RAN\)](#) leading to underutilization of precious radio resources and a sub-optimal user experience. An intelligent RAN analytics application residing within the network could pass on the radio channel information to the video server which would automatically assist the TCP congestion control decisions leading to better to user experience.

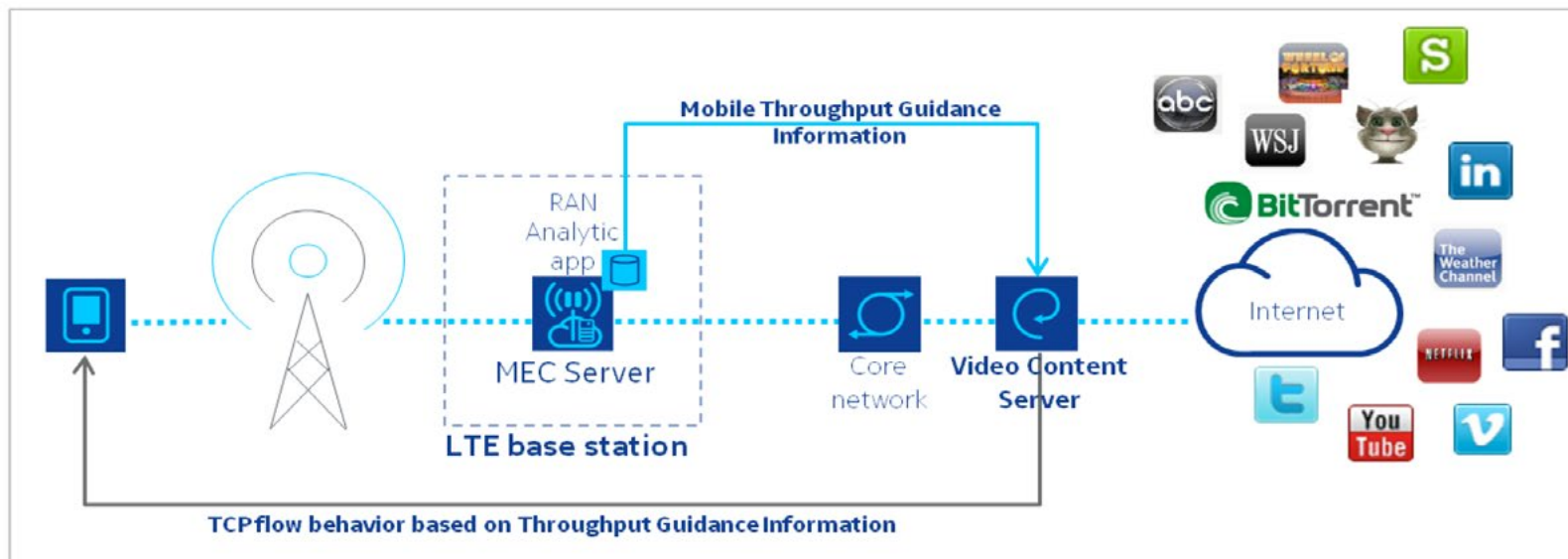


Figure 2: RAN aware video optimization | Source: ETSI



Similarly, factories churning out millions of data points a day may not want to put all the data on a public cloud for security reasons. It may also be wastage of bandwidth to send every bit of data to the cloud. It probably makes more sense for the bulk of the raw data to stay within the premises of the factory.

In regulated industries, such as finance and healthcare, data needs to be stored and analyzed on-premises to comply with local regulations. Branches of financial institution can find non-compliant transactions in real-time and stop them more quickly, compared to sending the data to a central data center. Using MEC, enterprises with sensitive digital assets can protect the security and integrity of their data by providing real-time security monitoring for traffic anomalies.

According to a research paper by Dell, the Edge Cloud can collect a large amount of contextual data about users, prevailing network conditions and consumer behavior that can be invaluable to operational support systems (OSS) and business support systems (BSS) applications. Sending all this data to the core may

be counterproductive and costly, chewing up the bandwidth. It is more economical to run analytics at the edge. Analytics may include a range of activities such as event correlation, big data applications, IoT data processing, video analytics, and machine learning.

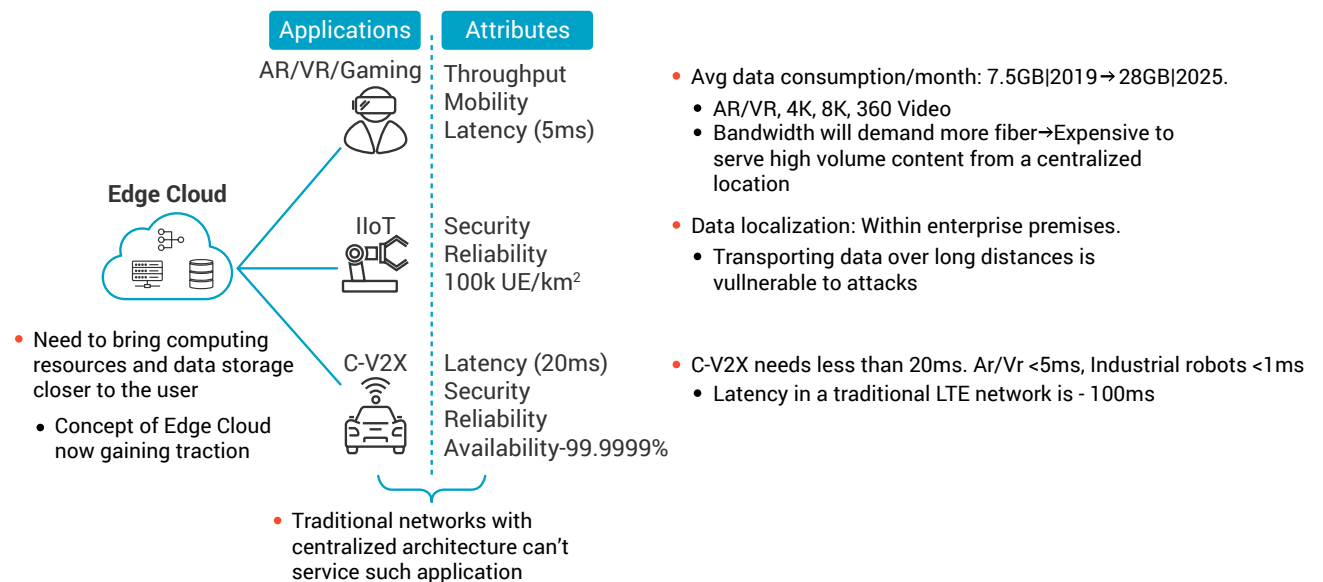


Figure 3: Application Landscape



From the above examples it is evident that there will be an increasing need for computation, storage, and data security closer to the end user (whether it is an individual or an enterprise). The Edge Cloud provides an IT environment with cloud computing capabilities within the telecom network and near the customer. This IT service environment will allow a multitude of applications from various providers including operators, content providers, application developers, etc. to run seamlessly.

Research firm [Gartner](#) claims that 75% of the data will be processed at the Edge by 2025. According to [Juniper Research](#) the Edge Cloud spend is forecasted to reach \$11b in 2024 from \$1.3B in 2019. Japanese operator [Rakuten](#) is planning to deploy 4000 edge sites in Japan by 2026. Since 2013, [EdgeConneX](#) has built more than 40 edge computing data centers across North America, Europe, and South America.



Edge Ecosystem

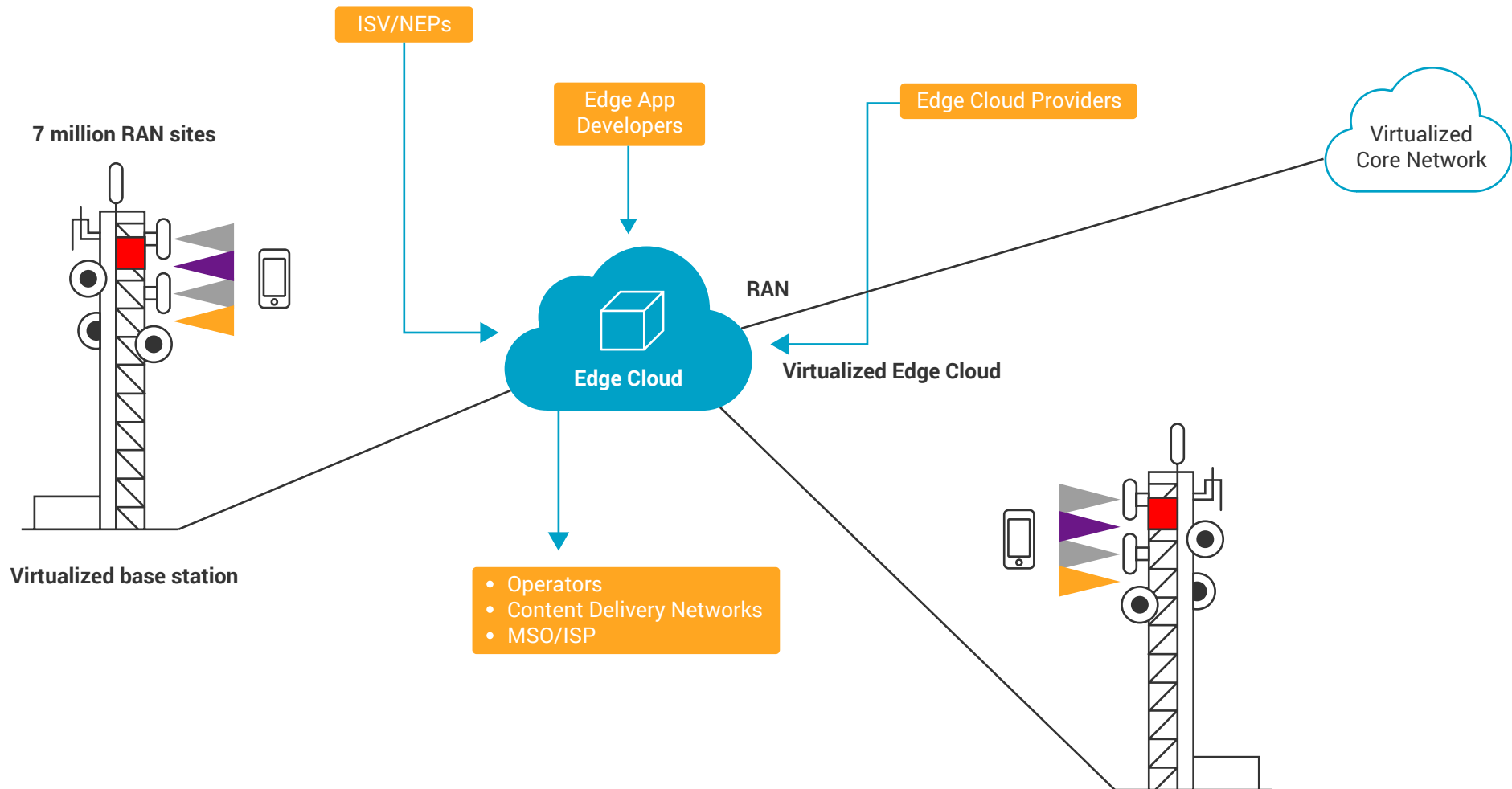


Figure 4: Edge Cloud Ecosystem



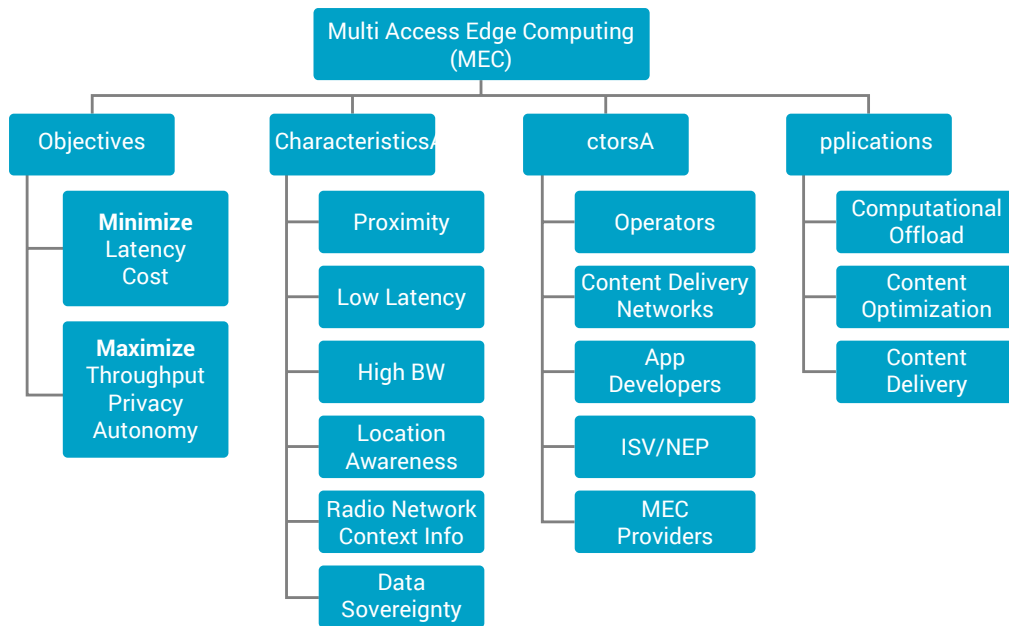


Figure 5: Taxonomy of Edge Computing

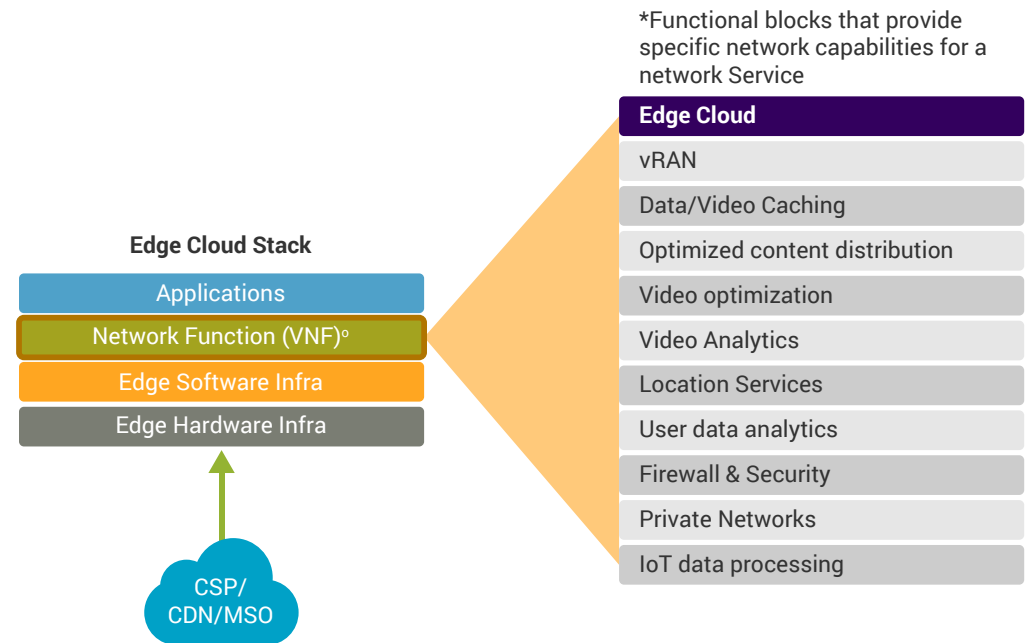


Figure 6: Edge Cloud - Technology Stack



a) MEC Reference Architecture

The ETSI MEC framework has three entities which are categorized into mobile edge system level, mobile edge host level, and networks level entities. The mobile edge host level is the fundamental part of the MEC framework. It consists of two main parts, namely the mobile edge host and the mobile edge host level management. The mobile edge host provides the virtualization infrastructure and the mobile edge platform, facilitating the execution of mobile edge applications.

The mobile edge platform enables applications to discover, advertise and consume edge services, and provides the virtualization infrastructure with a set of rules for the forwarding plane.

The underlying networks level offers connectivity to a variety of access technologies. On top, the mobile edge system level management provides an abstraction of the underlying MEC system facilitating access for UEs and third parties. [IEEE](#)

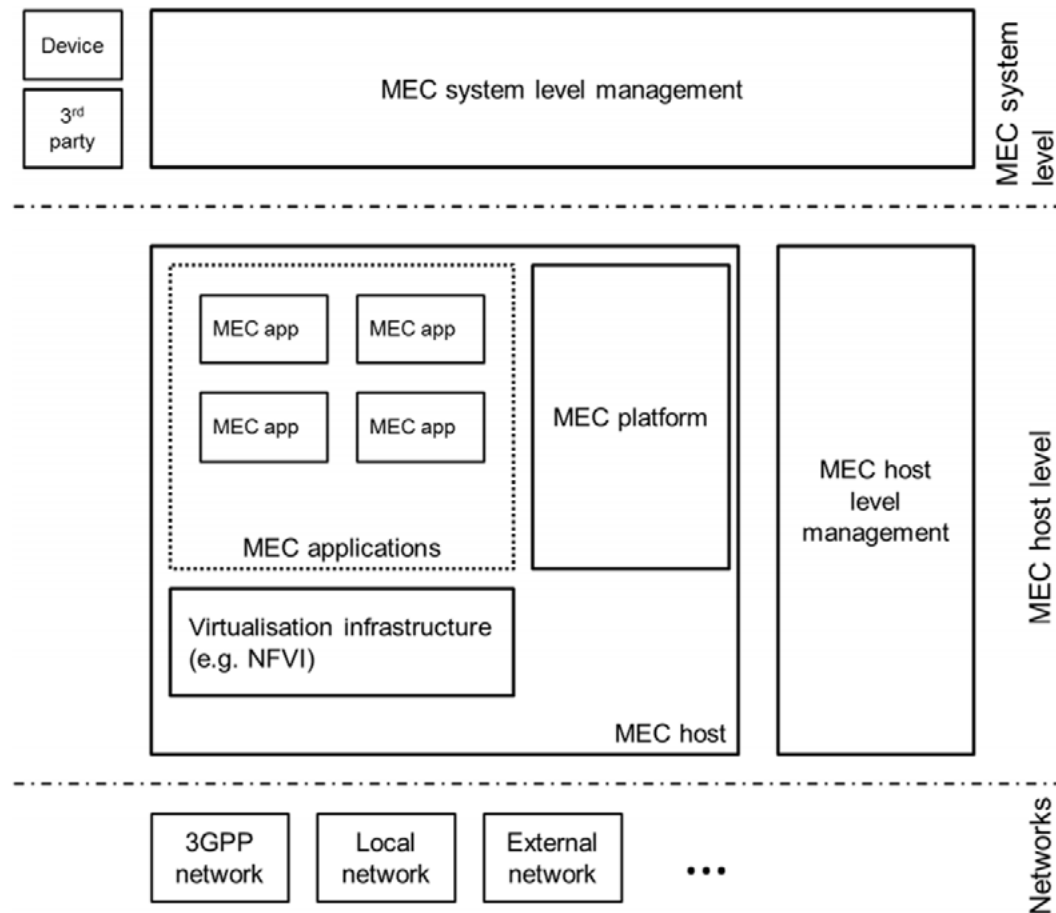


Figure 7: Multi-Access Edge Computing Framework | Source: [ETSI](#)



b) Network Slicing

A paper on MEC is incomplete without talking about network slicing. Traditional network design is one-size-fits-all. It is not possible to provide differentiated services using the same infrastructure. Using the traditional networks, if one were to offer low latency services like C-V2X, or a high reliability network for public safety or hospital, or a network for a smart city having thousands or millions of low power devices, then one would have to set up three different networks. Each of the mentioned use cases need different attributes of the network and hence will require dedicated infrastructure. But if the applications seldom use the infrastructure to its full capacity then it is a waste of resources and, not to mention, expensive. For example, a public safety network is not utilized all the time.

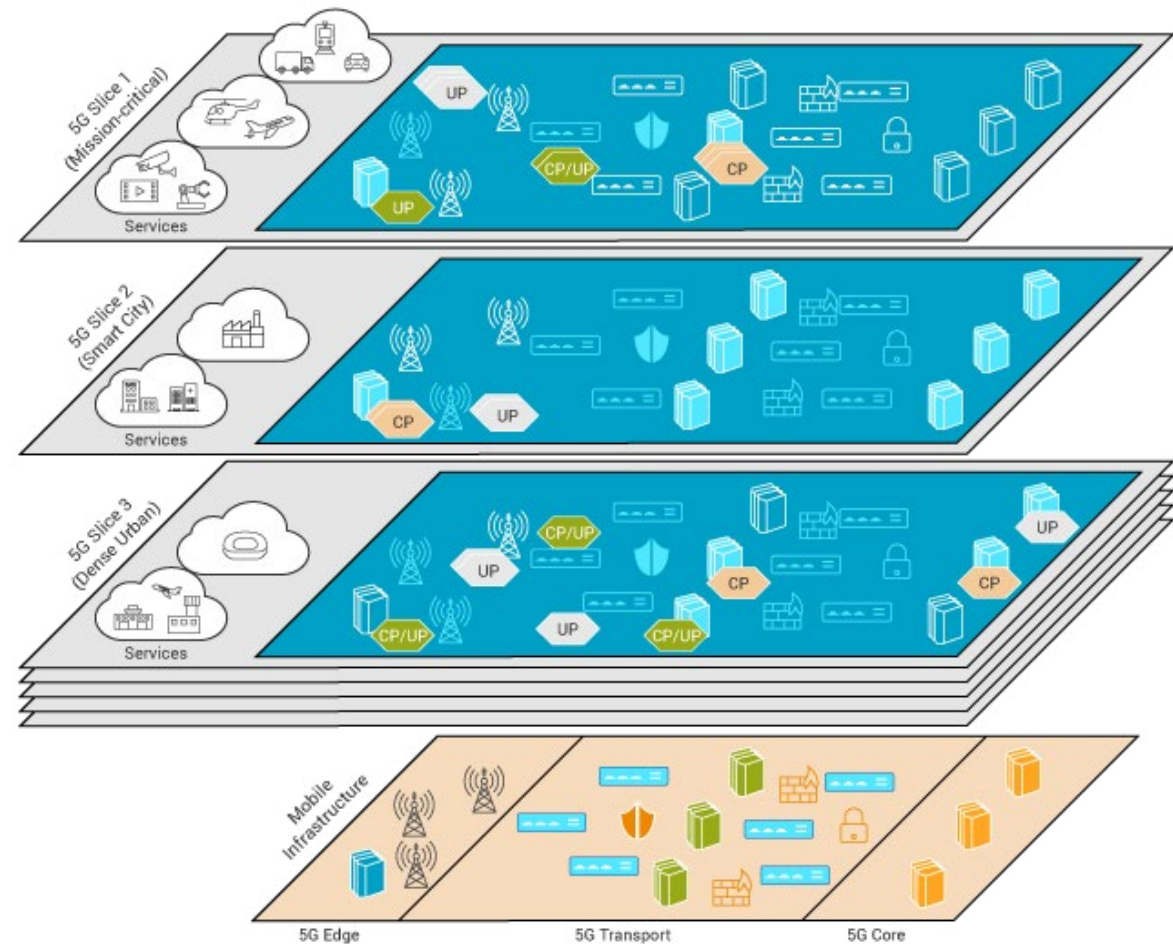


Figure 8: Network Slicing and role of MEC



Network slicing circumvents this problem by utilizing a common infrastructure to provide differentiated services. This reduces the capex for an operator. MEC is a catalyst for network slicing as the localized compute and storage resources can dynamically route the resources for the required application. This is of course helped by the Control plane and User plane Separation (CUPS) principle which is part of the 3GPP specification.

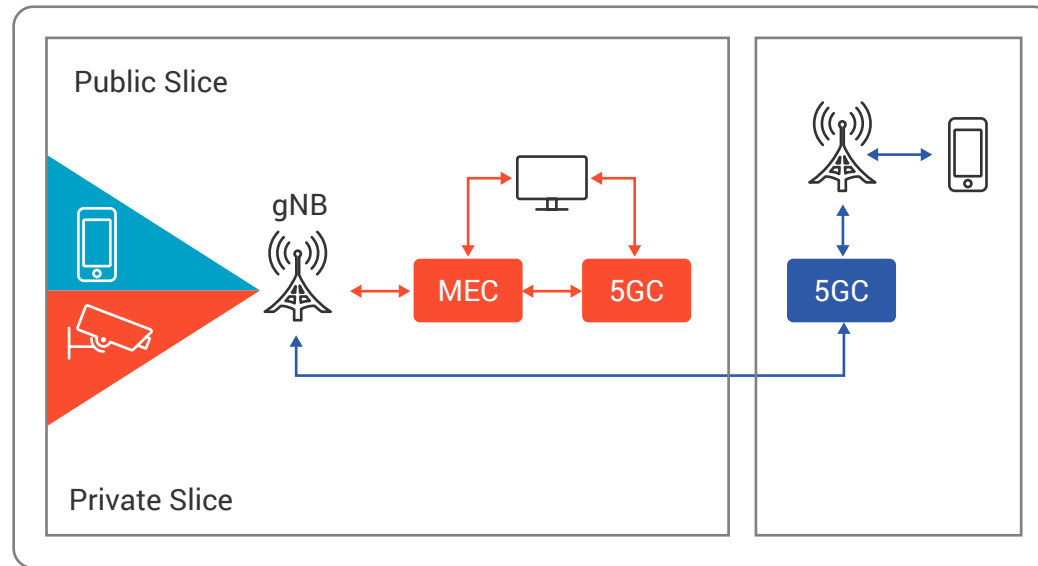


Figure 9: An Example of Network Slicing



How telecom operators can regain lost territory

Telecom operators are occupying a huge amount of real estate. Compared to the Cloud providers, they are in an advantageous position as the real estate they occupy is very close to customers in a vast majority of cases. Telecom companies can provide Edge infrastructure or platform services to Cloud providers and content delivery networks. The cost of setting up additional infrastructure for telecom operators is minimal if real estate is already available. In many instances, telecom operators can also

provide last mile fiber connectivity. Telecom companies can host several applications on its Edge infrastructure thus competing with the Cloud providers. There is a potential that telecom operators can also serve as content delivery networks in the markets they serve. Similarly, cable hub sites can be used for 5G Edge sites. The hub sites can be used for streaming content, server hosting, online gaming, etc.

Shaping the Edge Cloud

The numerous Virtual Network Functions (VNFs) and applications need to be integrated into the operators OSS and BSS systems in order to have a seamless service. The VNFs will have to be validated against the operator's infrastructure and onboarded to render the network service. The operator also needs to ensure that the applications conform to certain criteria like how much resources it consumes under different load and network conditions and whether an application has a malefic effect on other applications or VNFs. This is because the applications and the VNFs utilize the same infrastructure.

The operator will have to develop appropriate APIs (possibly 100s) for the applications and VNFs that run at the Edge site. Each Edge computing application will have different requirements. For example, in the smart manufacturing landscape, APIs and platforms

are bound to be different from those required for gaming or other entertainment-related use cases. The edge computing solution must be able to accommodate platforms for multiple solutions.

At the moment, there is no industry standard for the various API interfaces. As time progresses the operators' consortium could propose standardized APIs on the lines of the Open RAN interfaces. This should speed up the adoption of applications at the Edge. VNF onboarding and application validation need to be automated. The number of VNFs and applications will increase in the following years. An operator can have 100s or 1000s of Edge sites. Manually validating and deploying the applications is expensive and time consuming which will seriously retard the process of digitalization and thereby increase the time to revenue.



Cloud-native design principles has become a common design pattern for telecom workloads as this enables significant gains in terms of efficiency and innovation. Kubernetes seems to be the most-preferred platform for container-based, cloud-native applications in the telecom industry.

Dedicated runtime environments will be necessary for many industrial applications. The edge will generally need to host different runtime environments. As a result, efficient management of a multitude of different runtime environments is one of the most important capabilities of the edge-computing solution.

There are a multitude of APIs and runtimes because of multiple use cases. While some developers use a popular platform like Kubernetes or embrace web-scale platforms like Google or AWS, some verticals will probably develop their own platform and/or APIs. Examples of industry specific platforms are the 5G-ACIA (5G Alliance for Connected Industries and Automation) consortium and AECC (Automotive Edge Computing Consortium).

Telecom networks are getting denser with addition of more base stations and small cells. Added to it is the complexity due to heterogeneous technologies like 2G/3G/4G/5G and Wi-Fi. Managing such a network manually becomes extremely complex and expensive. Network automation and orchestration becomes imperative to contain costs and to bring about agility in network changes to render new services. Telecom companies are being challenged by OTT and Cloud service providers with their agility in launching new services. According to a [Cisco](#) study of its customers, CSPs can save 50-60% on opex by automating its network services. Embracing open network automation platforms like ONAP enables innovation across an ecosystem of partners and vendors as well as internal CSP departments.



Conclusion

It is apparent how Edge Computing will play a vital role in various applications irrespective of whether the network is 4G or 5G. Telecom operators, CDN providers, cable operators, cloud infrastructure providers are striving to have a presence at the Edge. Each of these players will deploy their Edge Cloud using different infrastructures (NFVI), different runtime environments and deploy VNFs and applications from different sources. The applications at the edge must be Cloud native to deliver high performance services and conserve constrained resources at the edge.

About Author

Pradeep has a decade of experience in the electronics and communications industry. At Sasken, he works for the Product Engineering Practice and is responsible for developing and marketing new services for the cellular technology industry.

The successful development of an edge-computing solution requires a good understanding of the use cases, associated deployment options and application-developer communities. For the ecosystem to flourish, it is important that third-party applications are able to work and interoperate on the edge sites.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





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